

EL-1005: Digital Logic Design Lab

Sections: CY – A , B

Serial No:

Sessional Exam

Total Time: 1 Hour 10 Min

Total Marks: 30

Lab Instructor

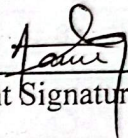
Imran Khan

Signature of Invigilator

Aqsa Noor
Student Name

24th-2010
Roll No.

CY-A
Course Section


Student Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Attempt on question paper. Attempt all of them. Read the question carefully, understand the question, and then attempt it.
2. No additional sheet will be provided for rough work. Use the back of the last page for rough work.
3. If you need more space, write on the back side of the paper and clearly mark question and part number etc.
4. Calculator sharing is strictly prohibited.
5. Use permanent ink pens only. Any part done using soft pencil will not be marked and cannot be claimed for rechecking.

+1

	Q-1	Q-2	Q-3	Total
Marks Obtained	10	9.5	7.5	28
Total Marks	10	10	10	30

Excellent

Question 1 [10 Marks]

10/1

You must implement a circuit that has three inputs: A, B, and C, and one output: F. The output F should be 1 only when an even number of inputs are HIGH, except when all inputs are LOW (in which case F must be 0).

You must use a combination of universal gates (NAND or NOR).

No other gates are allowed.

Use Proteus to implement the circuit and verify its functionality.

You will get bonus marks if you use only NAND gate.

Truth Table:

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Circuit expression

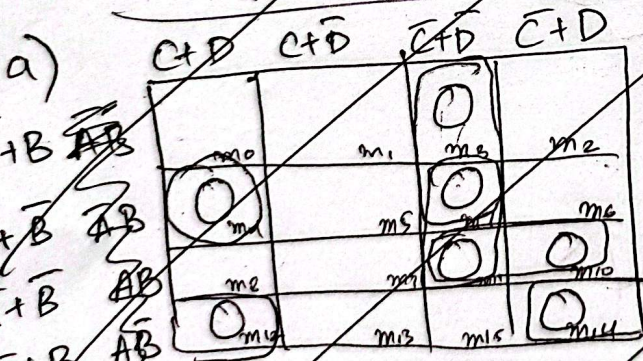
$$\Rightarrow \bar{A}BC + A\bar{B}C + AB\bar{C}$$

Made using NAND gates only

Question 2 [10 Marks]

Simplify the following Boolean function using a 4-variable Karnaugh Map and express your final answer in Simplified Product of Sums (POS) form.

- $f(A,B,C,D) = \prod M(3,4,7,10,11,12,14)$
- $f(A,B,C,D) = \prod M(0,1,2,5,6,8,9)$



$$\Rightarrow \cancel{(A+B)} \cancel{(C+D)} (A+\bar{B}+C+D) (\bar{A}+B+D) (\bar{A}+\bar{B}+\bar{C}+\bar{D})$$

$$\Rightarrow \text{Simplified form}$$

$$\Rightarrow (A+\bar{B}+C+D) (\bar{A}+B+D) (\bar{A}+\bar{B}+\bar{C}) (A+\bar{C}+\bar{D})$$

9.5

0072

$$b) f(A, B, C, D) = \pi M(0, 1, 2, 5, 6, 8, 9)$$

	$C+D$	$C+\bar{D}$	$\bar{C}+\bar{D}$	$\bar{C}+D$
$A+B$	$\bigcirc_{M_0}$	$\bigcirc_{M_1}$	M_2	$\bigcirc_{M_3}$
$A+\bar{B}$	M_4	$\bigcirc_{M_5}$	M_6	$\bigcirc_{M_7}$
$\bar{A}+\bar{B}$	M_8	M_{10}	M_{12}	M_{14}
$\bar{A}+B$	$\bigcirc_{M_9}$	$\bigcirc_{M_{11}}$	M_{13}	M_{15}

$$\Rightarrow (B+C)(A+C+\bar{D})(A+\bar{C}+D)$$

$$a) f(A, B, C, D) = \pi M(3, 4, 7, 10, 11, 12, 14)$$

	$C+D$	$C+\bar{D}$	$\bar{C}+\bar{D}$	$\bar{C}+D$
$A+B$	M_0	M_1	$\bigcirc_{M_2}$	M_3
$A+\bar{B}$	$\bigcirc_{M_4}$	M_5	$\bigcirc_{M_6}$	M_7
$\bar{A}+\bar{B}$	$\bigcirc_{M_8}$	M_{10}	M_{12}	$\bigcirc_{M_{14}}$
$\bar{A}+B$	M_9	M_{11}	$\bigcirc_{M_{13}}$	$\bigcirc_{M_{15}}$

$$\Rightarrow (A+\bar{C}+\bar{D})(\bar{B}+C+D)(\bar{A}+B+\bar{C})(\bar{A}+\bar{C}+D)$$

Circuit on Proteus

75

Question 3 [10 Marks]

You are given a combinational circuit with inputs A, B, Cin, and a Mode control M, where:

When $M = 0$, the circuit must behave as a Half Adder.

When $M = 1$, it must behave as a Full Adder.

(a) Derive the Boolean expressions for the merged Sum and Carry outputs in terms of A, B, Cin, and M.

(b) Simplify both expressions using K-Maps.

(c) Implement the final expressions using only XOR (7486), AND (7408), OR (7432), and NOT (7404) gates.

$M = 0$

$$\text{Sum} = A \oplus B$$

$$\text{Carry} = AB$$

$M = 1$

$$\text{Sum} = A \oplus B \oplus C$$

$$\text{Carry} = C(A \oplus B) + AB$$

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Rough

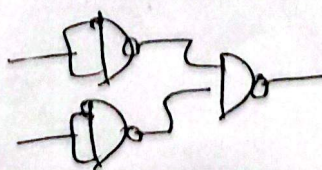
	$\bar{A}\bar{B}$	$\bar{A}B$	$A\bar{B}$	AB
$\bar{A}$			1	
A		1		1

$$C(\bar{A}B + A\bar{B}) + AB$$

$$(A \oplus B)C + AB$$

$$AB + BC + AC$$

$$\Rightarrow C(A \oplus B) + AB$$



A	B	C	D
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1